JOSHUA OFFE BERKOH

Threat Intelligence and Security Research | Threat Hunting | Dark-Web Intelligence

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PROFESSIONAL SUMMARY

Cybersecurity professional and PhD researcher focused on cyber threat investigations, threat hunting, OSINT, security operations, and dark-web intelligence research. Publishes public, scenario-based threat investigation reports through the Cyber Threat Intelligence Lab, applying KQL, Azure Data Explorer, IOC pivoting, timeline reconstruction, and MITRE ATT&CK mapping. Combines SOC experience, security engineering, vulnerability research, and Python-driven research workflows to produce defensible intelligence and hidden-service ecosystem analysis.

CORE COMPETENCIES

Threat Investigations	KQL log analysis, Azure Data Explorer, telemetry correlation, intrusion timeline reconstruction, IOC pivoting
Cyber Threat Intelligence	Intelligence collection, source evaluation, ATT&CK mapping, indicator analysis, structured reporting, confidence-based assessment
Threat Hunting and SOC	Hypothesis-driven hunting, alert triage, SIEM monitoring, incident response support, network traffic analysis
Dark-Web Research	I2P hidden-service discovery, infrastructure mapping, ecosystem analysis, graph analysis, Python and MariaDB workflows
Technical Tools	Python, SQL, KQL, Wireshark, Splunk, Burp Suite, Nmap, Metasploit, Nessus, Docker, Kali Linux, MariaDB

PUBLIC THREAT INVESTIGATION PORTFOLIO

Cyber Threat Intelligence Lab | Scenario-Based Threat Investigations **2026 - Present**

- Published professional investigation reports using KC7 realistic enterprise scenarios, clearly marked as training-environment investigations rather than client incidents.
- Applied KQL and Azure Data Explorer to analyze email, process, authentication, network flow, file creation, passive DNS, and web telemetry.
- Documented incident timelines, IOCs, confidence levels, observed TTPs, MITRE ATT&CK mappings, investigative conclusions, and evidence screenshots.
- Solvi Systems: A Tale of Supply Chains and ICS. Reconstructed a critical-infrastructure supply-chain intrusion affecting an ICS software vendor, from web reconnaissance and spear phishing to C2 persistence, lateral movement, and SDLC source-code exfiltration. Identified 470 persistent C2 connections across 38 endpoints and mapped 8 ATT&CK techniques.
- Inside Encryptodera: An Insider Threat Scenario. Analyzed a dual-track financial services compromise involving 27-day FTP exfiltration of cold-storage wallet data and hijacked-identity Active Directory ransomware. Correlated network, file, process, and authentication telemetry; documented 306 ransomware-impacted endpoints and mapped 8 ATT&CK techniques.
- Valdoria Votes. Investigating a public-sector election-infrastructure APT scenario focused on persistence mechanisms, multi-hop C2, domain registrar anomalies, and infrastructure tracking. Status: in progress.

PROFESSIONAL EXPERIENCE

Security Engineer Intern | Intuit Inc. | San Diego, CA **Jun 2023 - Sep 2023**

- Consolidated security repositories into red-team toolsets, improving operational efficiency by 20%.
- Automated code-based compliance validation through Tox integration, reducing manual review effort by 30%.
- Resolved compliance-related security issues before deployment, contributing to remediation of 70% of identified vulnerabilities.
- Collaborated with cross-functional teams to improve security controls, deployment readiness, and control validation workflows.

Security Operations Center Analyst | Virtual Infosec Africa | Accra, Ghana **Oct 2021 - Jun 2022**

- Monitored and analyzed network activity supporting a 95% intrusion detection rate for a major financial institution.
- Investigated security events and supported incident response activities, reducing response times by 30%.
- Assisted with deployment and operation of SOC security solutions that contributed to a 20% reduction in cybersecurity incidents.
- Performed threat monitoring, alert triage, and suspicious activity analysis to support proactive mitigation.

Adjunct Instructor | University of Cincinnati | Cincinnati, OH **Jan 2023 - Apr 2023**

- Delivered hands-on cybersecurity learning activities focused on practical security concepts, applied investigation, and problem solving.
- Improved student performance by 15% through applied exercises, structured feedback, and technical mentoring.

RESEARCH AND TECHNICAL WRITING

I2P Hidden Service Ecosystem Analysis | PhD Research, University of Cincinnati **2025 - Present**

- Design and operate a Python-based collection and analysis framework for studying hidden services within the I2P anonymity network.
- Collect and structure application-layer observations from eepsite crawls and related network observations using MariaDB and reproducible pipelines.
- Analyze hidden-service connectivity, infrastructure relationships, and ecosystem behavior through graph-based workflows.
- Frame the work as dark-web intelligence research focused on infrastructure and ecosystem measurement, not real-world adversary attribution.

Threat Analysis, Detection and Mitigation | University of Cincinnati **Jan 2023 - Apr 2023**

- Researched APT tactics, techniques, procedures, and defensive countermeasures.
- Applied behavioral analytics and threat intelligence concepts to support detection and mitigation planning.

VULNERABILITY RESEARCH AND OSINT EXPERIENCE

Bug Bounty Hunter | Bugcrowd | Remote **Jan 2021 - Dec 2021**

- Identified and responsibly disclosed vulnerabilities across seven programs.
- Earned Hall of Fame recognition from multiple organizations, including Centrify, Arlo, and Humble Bundle.

Security Competitions and Practical Assessments | Hacker101, MetaCTF, AppSec Challenges | Remote **2021 - Present**

- Completed practical web security, forensics, reconnaissance, cryptography, and exploitation challenges.
- Identified SQL injection, cross-site scripting, and server-side request forgery vulnerabilities through application security challenges.

TraceLabs | OSINT CTF Player and Judge | Remote **Jan 2021 - Dec 2022**

- Applied OSINT techniques to identify critical leads in missing-persons cases and provided judging feedback for CTF competitions.

EDUCATION

University of Cincinnati | PhD, Information Technology | Expected April 2928
University of Cincinnati | Master of Science, Information Technology | Aug 2024

PROFESSIONAL DEVELOPMENT AND COMMUNITY

Practical Detection Engineering Study, in progress | KC7 Cyber Security Analyst Modules | OWASP Cincinnati Bugbash Mentor | ISC2 Examination Developer | AWS Community Builder, Security Division